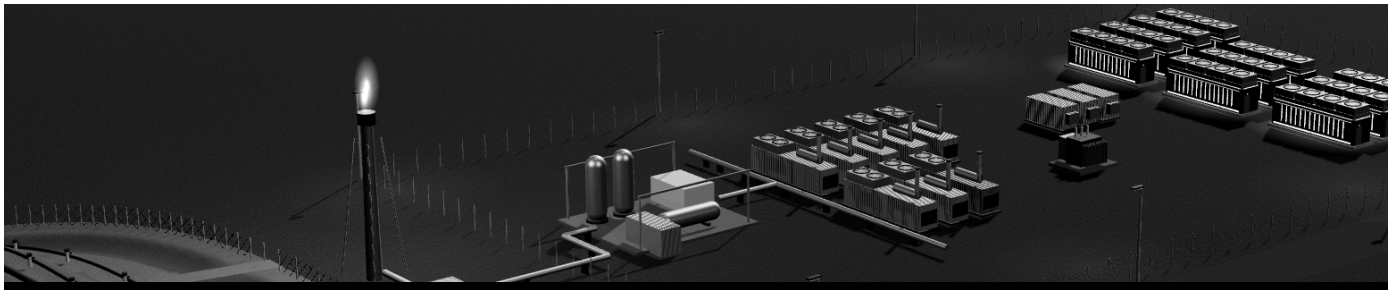
Modern Landfill, York, Pennsylvania: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

10.6 MW	10	18 ac	971k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $2.91M	$18k to $35k	$18k to $2.95M



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=4272&CH4=44.7

THE MATH

Landfill gas collected (2023)	4,272 scfm
Methane content (2023)	44.7%
Energy in the gas	116.7 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 10.6 MW
Compute (power less cooling and losses, /1.2)	about 8.8 MW
Compute modules (about 1 MW each)	10
Gas engines (running plus one spare)	8 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 18 acres
Gas bought per year (95% run time)	971k MMBtu
Gas payment, up to $3.00/MMBtu	up to $2.91M
Land lease at $1,000 to $2,000 per acre per year	$18k to $35k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Whether any of the gas is set aside for a new project
- Gas outlook as the landfill nears capacity
- Pennsylvania DEP air plan approval for the engines
- Space for a fenced parcel near the flare

Source: EPA Greenhouse Gas Reporting Program (GHGRP): about 4,272 scfm collected in 2023 at 44.7% methane (4,582 to 5,013 scfm in 2019-2022). EPA LMOP lists the 2022 flow as flared; the 9.1 MW engine plant there is listed as shut down in 2013. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.